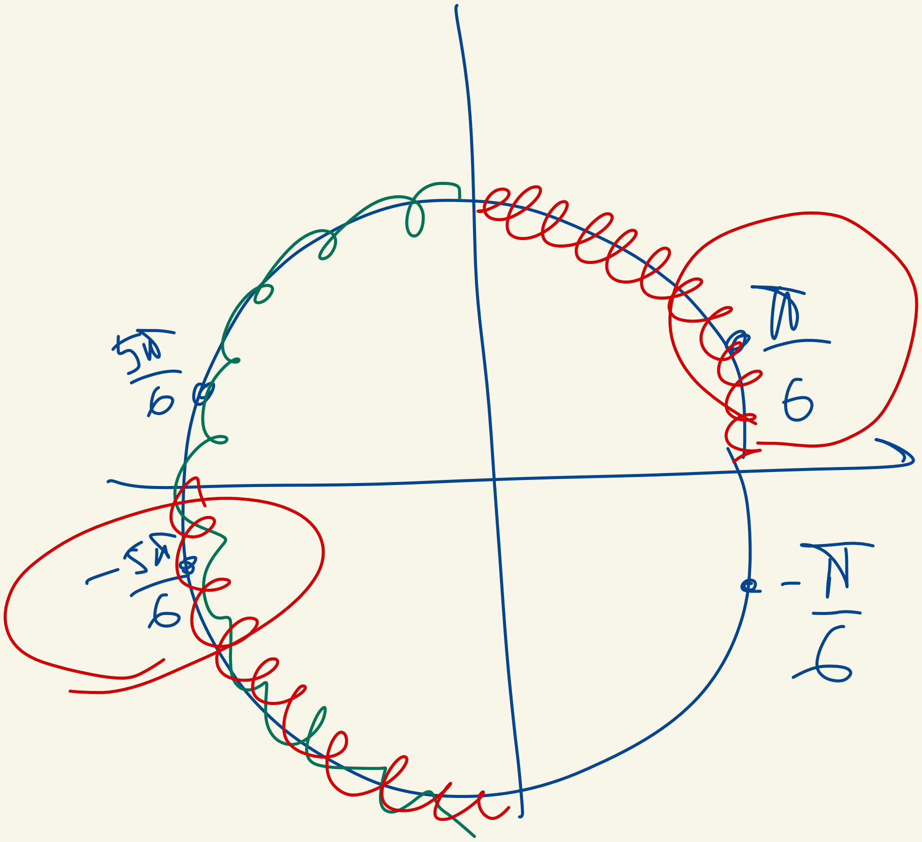


$$x_1 = \frac{\pi}{6} + 2\pi k$$

$$x_2 = -\frac{5\pi}{6} + 2\pi k$$

$$\frac{1}{x(x+1)} + \frac{1}{(x+1)(x+2)} + \frac{1}{(x+2)(x+3)} \leq \frac{3}{4}$$

$$\ldots$$

$$\rightarrow \frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$$

$$\frac{(n+1)-n}{n(n+1)} = \frac{1}{n(n+1)}$$

$$\frac{1}{x} - \cancel{\frac{1}{x+1}} + \cancel{\frac{1}{x+1}} - \cancel{\frac{1}{x+2}} + \cancel{\frac{1}{x+2}} - \frac{1}{x+3} \le \frac{3}{4}$$

$x \neq -1$

$x \neq -2$

$x \neq 0$

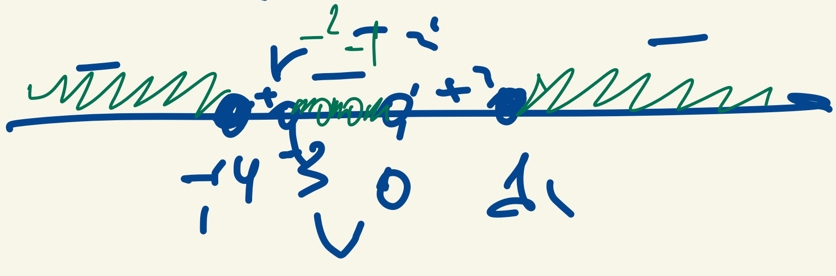
$x \neq -3$

$$\Rightarrow \frac{1}{x} - \frac{1}{x+3} \le \frac{3}{4}$$

$$\frac{x+3-x}{x(x+3)} \le \frac{3}{4}$$

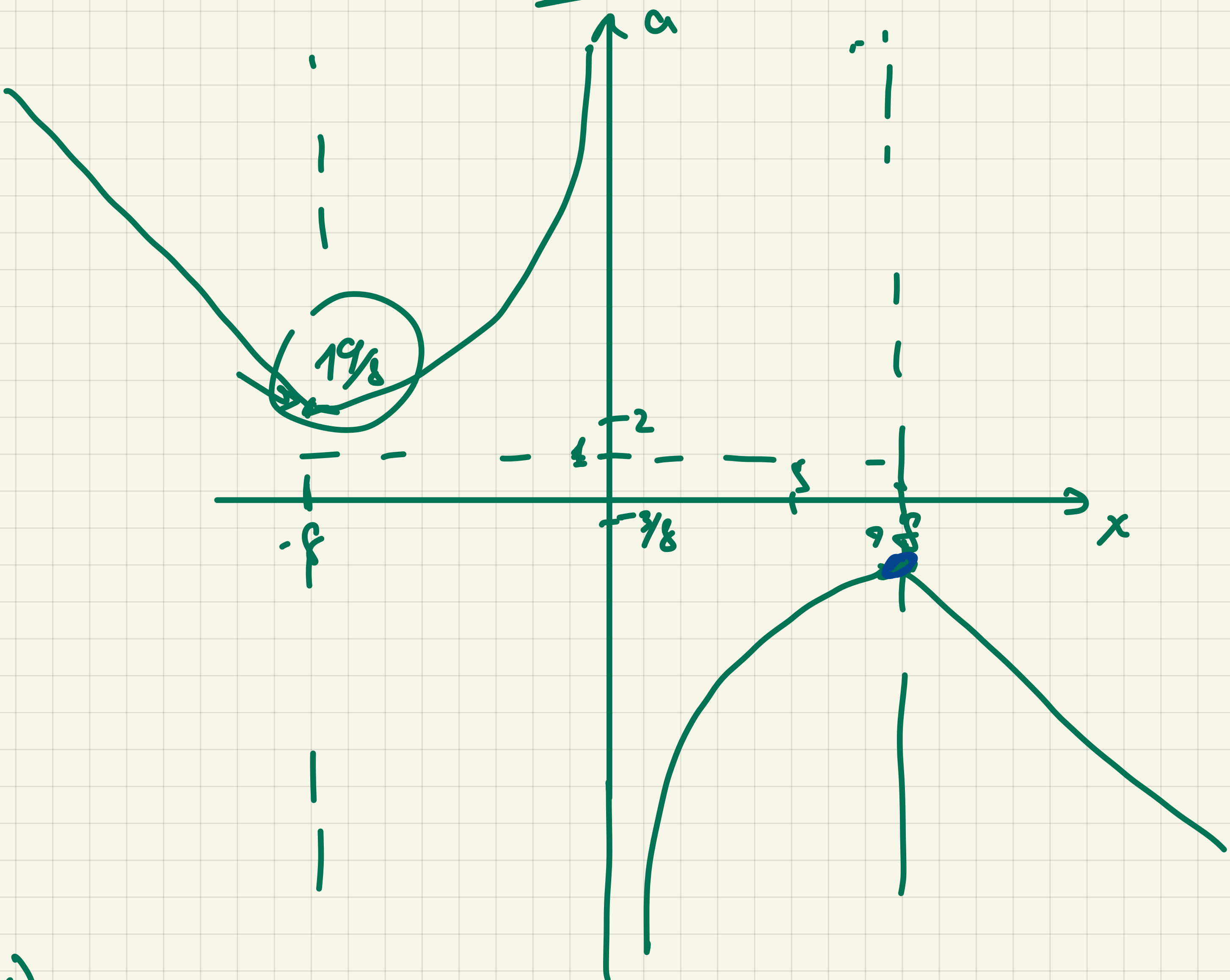
$$\frac{3 \cdot 4 - 3(x)(x+3)}{4(x+3)x} \le 0$$

$$\frac{12 - 3x^2 - 9x}{4x(x+3)} \le 0$$



$$x \in (-\infty; -4] \cup (-3; -2) \cup (-2; -1) \cup (-1; 0) \cup [1; +\infty)$$

$$a|x+8|+(2-a)\underline{|x-8|}+6=0$$



1) $x>8$

$$ax+8a+(2-a)(x-8)+6=0$$

$$\cancel{ax}+8a+2x-16-\cancel{ax}+8a+6=0$$

$$16a-10+2x=0$$

$$a=-\frac{2x+10}{16}=\frac{5-x}{8}$$

2) $a(x+8)+(2-a)(-x+8)+6=0$

$$ax + \cancel{8a} - 2x + 16 + ax - \cancel{98} + 6 = 0$$

$$2ax - 2x + 22 = 0$$

$$2ax = 2x - 22$$

$$ax = x - 11$$

$$a = 1 - \frac{11}{x} \Rightarrow$$

$$x = 8 \quad a = 1 - \frac{11}{8} = \frac{8-11}{8} = -\frac{3}{8}$$

$$x = -8 \quad a = 1 + \frac{11}{8} = \frac{19}{8}$$

3) $$-ax - 8a + (2-a)(-x+8) + 6 = 0$$

$$-\cancel{ax} - 8a - 2x + 16 + \cancel{ax} - 8a + 6 = 0$$

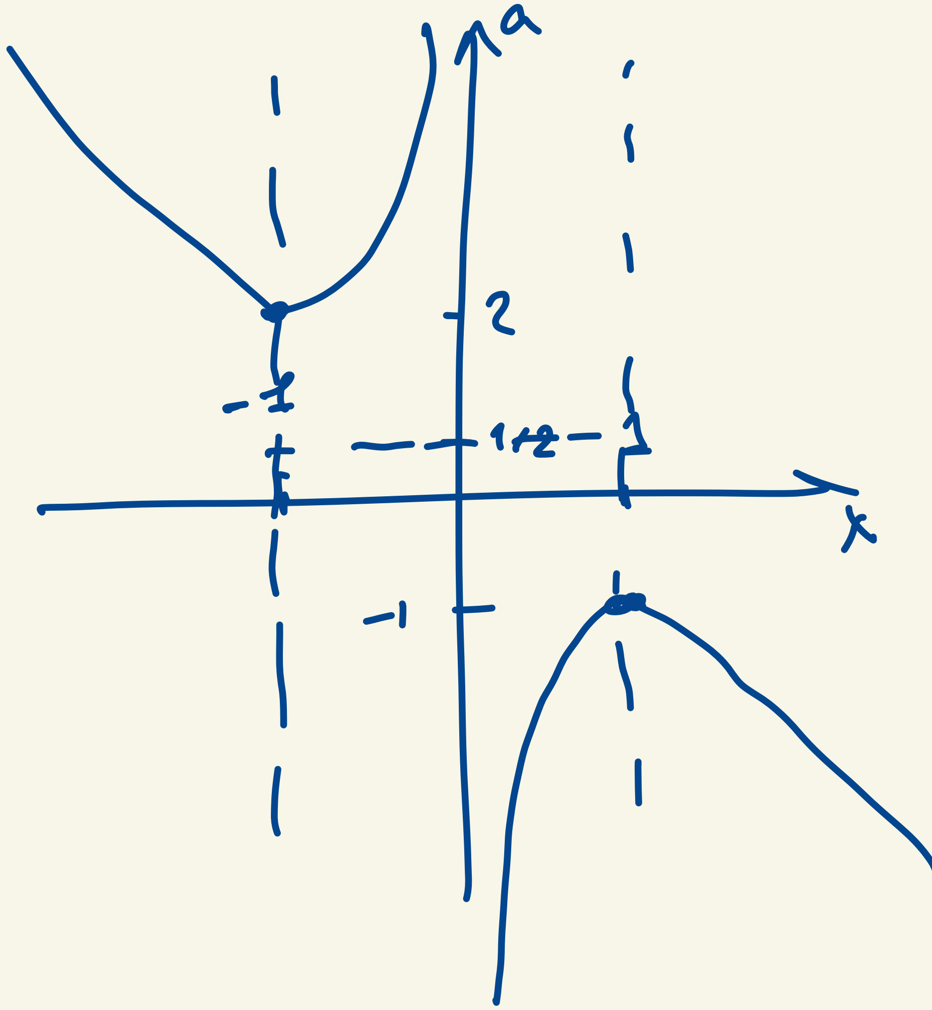
$$-16a - 2x + 22 = 0$$

$$a = \frac{22 - 2x}{16} = \frac{11 - x}{8}$$

$$x = -8$$

$$a = \frac{11 + 8}{8} = \frac{19}{8}$$

$$a|x+1| + (1-a)|x-1| + 2 = 0$$



$x > 1$

$$a = \frac{-x-1}{2}$$

$$a = \frac{x-3}{2x} =$$

$$\frac{1}{2} - \frac{3}{2x}$$

$a(1) = -1$

$a(-1) = 2$

$a < -1$

$x = -1$

$$-ax - a + (1-a)(-x+1) + 2 = 0$$

$$-\cancel{ax} - a - x + 1 + \cancel{ax} - a + 2 = 0$$

$$-2a - x + 3 = 0$$

$$\frac{-x+3}{2} = a$$

$a(-1) = 2$

$$|x+a^2| = |a+x^2|$$

> 3 реш

1. $(\quad)^2 = (\quad)^2$

$$(x+a^2)^2 = (a+x^2)^2$$

$$(x+a^2)^2 - (a+x^2)^2 = 0$$

$$(x+a^2-a-x^2)(x+a^2+a+x^2) = 0$$

$$\left[\begin{array}{l} x+a^2-a-x^2=0 \\ x+a^2+a+x^2=0 \end{array}\right.$$

$$\underline{x^2-x-a^2+a} = \underline{x^2-x}+a(1-a)$$

$$\left(x^2-x+\frac{1}{4}\right)-\frac{1}{4} \qquad \underbrace{\left(x-\frac{1}{2}\right)^2-\frac{1}{4}}$$

$$\left(x-\frac{1}{2}\right)^2-\left(a+\frac{1}{2}\right)^2=\frac{1}{2}$$

$$\left(x-\frac{1}{2}\right)^2-\frac{1}{4}-\left(a^2-a+\frac{1}{4}-\frac{1}{4}\right)$$

$$\left(x-\frac{1}{2}\right)^2-\cancel{\frac{1}{4}}-\left(a-\frac{1}{2}\right)^2+\cancel{\frac{1}{4}}=0$$

$$\left(x-\frac{1}{2}\right)^2-\left(a-\frac{1}{2}\right)^2=0$$

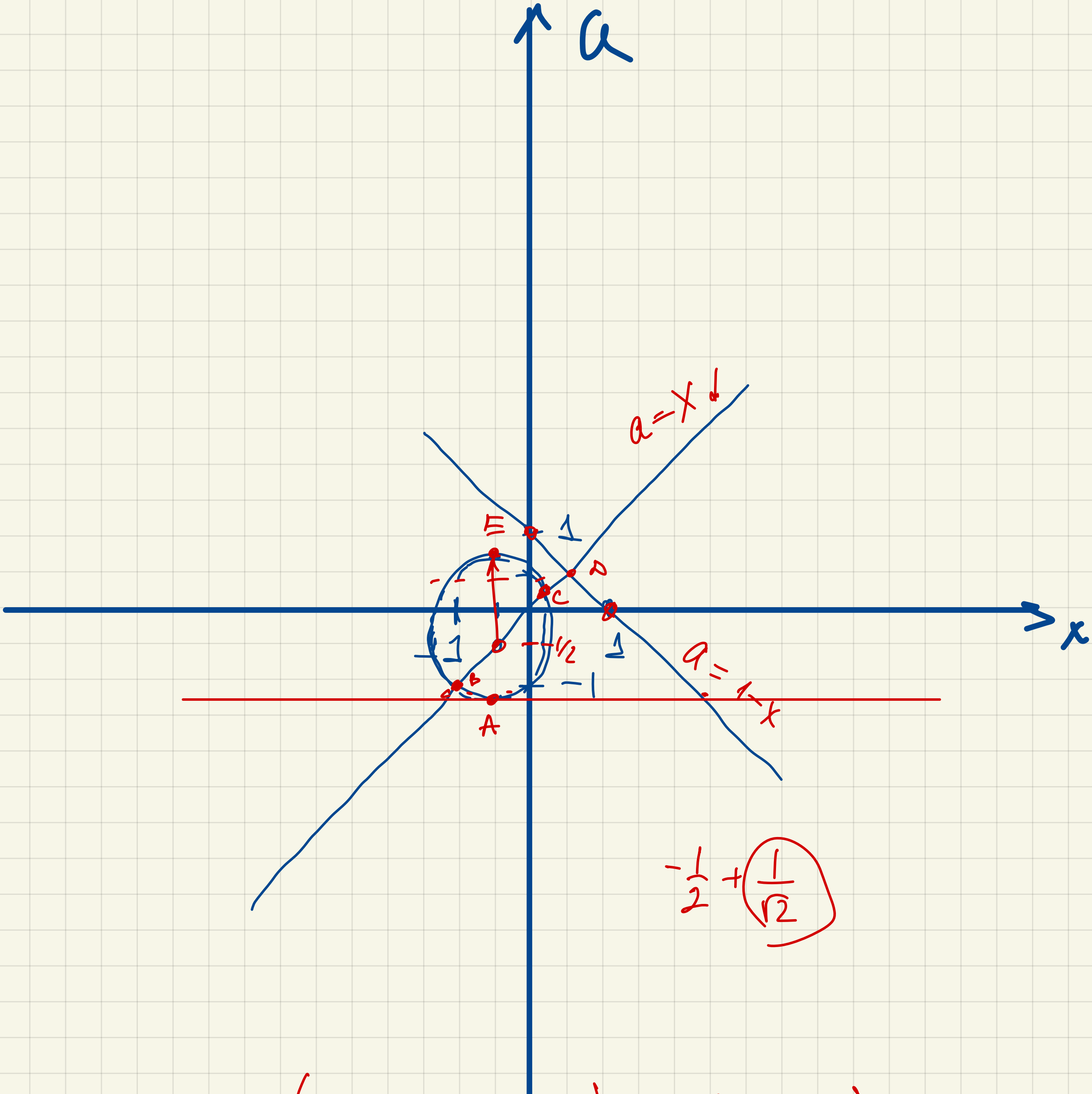
$$\left(x-\cancel{\frac{1}{2}}-a+\cancel{\frac{1}{2}}\right)\left(x-\frac{1}{2}+a-\frac{1}{2}\right)=0$$

$$\left[\begin{array}{l} x=a \\ x+a-1=0 \\ \underline{x^2+x+a^2+a=0} \end{array}\right.$$

$a=1-x$

$$\left(x+\frac{1}{2}\right)^2+\left(a+\frac{1}{2}\right)^2=\frac{1}{2}=R^2$$

$R=\frac{1}{\sqrt{2}}$



$-\frac{1}{2}+\frac{1}{\sqrt{2}}$

Отв: $\left(-\frac{1}{2}-\frac{1}{\sqrt{2}}; -1\right) \cup (-1; 0) \cup \left(0; \frac{1}{2}\right)$
$\cup \left(\frac{1}{2}; -\frac{1}{2}+\frac{1}{\sqrt{2}}\right)$

$$|x^2-2ax+7| = |6a-x^2-2x-1|$$

> 2 реш

$$\underbrace{(-2ax+6a-2x+6)}(2x^2-2ax-6a+2x+8)=0$$

$$\left[\begin{array}{l} 6a-2ax+6-2x=0 \\ 2x^2-2ax-6a+2x+8=0 \end{array}\right.$$

$$2a(3-x)+2(3-x)$$

$$\left[\begin{array}{l} (2a+2)(3-x)=0 \\ 2x^2+2x+8=a(2x+6) \\ a=\dfrac{2x^2+2x+8}{2x+6} \end{array}\right.$$